Hong Kong Metropolitan University

COMP8090SEF
Data Structures and Algorithms

Course Project Report - Task 1 (OOP-based Application Development)

Submission date: 17 April 2026

Academic honesty declaration (individual version):

I declare that:

(i) I have read and checked that all parts of the project (including proposal, code programs, and reports), they are contributed by me, here submitted is original except for source material explicitly acknowledged;

(ii) the project, in parts or in whole, has not been submitted for more than one purpose without declaration;

(iii) I am aware of the University's policy and regulations on honesty in academic work and understand the possible consequence when breaching such policy and regulations;

(iv) I confirm that I have declared in the report about the usage of AI and other generative models, including but not limited to ChatGPT, LLaMA, Gemini, Mistral, and Stable Diffusion, and complied with the instructions provided by HKMU; and

(v)I am aware that I should be held responsible and liable to disciplinary actions, irrespective of whether I have signed the declaration and whether I have contributed, directly or indirectly, to the problematic contents.

I confirm that I have read through and understood the project requirements. I understand that failure to comply with the project requirements will result in score deduction.

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GitHub repository: https://github.com/Dianthos/comp8090-yan-chudong-course-project

1. Problem Definition & Motivation

Students often juggle multiple courses, assignments, quizzes, and exams. Due dates are scattered across emails and LMS pages, which increases the risk of missing deadlines. This project builds a lightweight Student Assignment Manager to help track tasks by course and due date.

2. Solution Overview

The system provides the following core functions:

- Add tasks (Assignment / Exam / General)
- List all tasks with status, due date, and details
- Search and filter tasks by keyword and course
- Sort tasks by due date
- Mark tasks as completed / uncompleted
- Persist data to a local JSON file

3. OOP Concepts Used

The implementation demonstrates common OOP concepts as follows:

OOP Concept	Where it appears in the project
Class & Object	Task, AssignmentTask, ExamTask, GeneralTask, TaskManager, JsonStorage
Encapsulation	TaskManager stores tasks internally and exposes safe methods
Inheritance	AssignmentTask / ExamTask / GeneralTask inherit from Task
Polymorphism	details() and kind() behave differently by subclass
Abstraction	Task is an abstract base class with required methods
Modular programming	models.py, manager.py, storage.py, main.py

(See Appendix A for sample test cases.)

4. External Resources Declaration

External resources used:
- Python standard library: datetime, json, dataclasses, abc, typing

- Third-party packages/libraries: none in the submitted code
- AI assistance: ChatGPT was used to help brainstorm the project structure, refine report wording, and troubleshoot code issues. All code and report contents were reviewed, tested, and revised by the student before submission.
- Other references: please add any websites, screenshots, or tutorial sources you actually used.

All non-original materials should be cited in the References section.

5. Self-Reflection & Future Improvements

- Add a GUI (web or Tkinter) and calendar-style view
- Add reminders/notifications
- Support recurring tasks and priorities
- Add unit tests and stronger input validation

References

[1] COMP2090SEF/8090SEF Course Project Document (Version 1.7), Spring 2026.

[2] Lecture Note 2: Object-Oriented Programming (OOP) and Python Class.

[3] Python 3 Documentation, https://docs.python.org/3/.

Appendix A — Demonstration / Test Cases

Selected functional test cases are summarized below. These cases show the main consolebased features used during validation.

```
PS D:\OneDrive - Hong Kong Metropolitan University\桌面\COMP8090_Project_AssignmentManager> python -m task1.main
=== Student Assignment Manager ===

Menu
1) View all tasks
2) Add a task
3) Search
4) Filter by course
5) Sort by due date
6) Toggle done
7) Delete task
8) Save & exit
> 1

ID  | Done | Due        | Course        | Kind       | Title (details)
----+------+------------+---------------+------------+-----------------------------
1   |      | 2222-12-22 | 2222          | assignment | 222                 (weight=22%)

> 2

Add a task
1) Assignment  2) Exam  3) General
> 1
Title: Lab22
Course (e.g., COMP2090): COMP2222
Due date (YYYY-MM-DD): 2222-12-22
Points (optional): 22
✅ Added.
```

```
> 3
Keyword (title/course): COMP2222

Matches:
- [2] Lab22 (COMP2222) due 2222-12-22
```

```
> 1

ID  | Done | Due        | Course        | Kind       | Title (details)
----+------+------------+---------------+------------+-----------------------------
1   |      | 2222-12-22 | 2222          | assignment | 222                 (weight=22%)
2   |  ✅  | 2222-12-22 | COMP2222      | assignment | Lab22               (weight=22%)
```